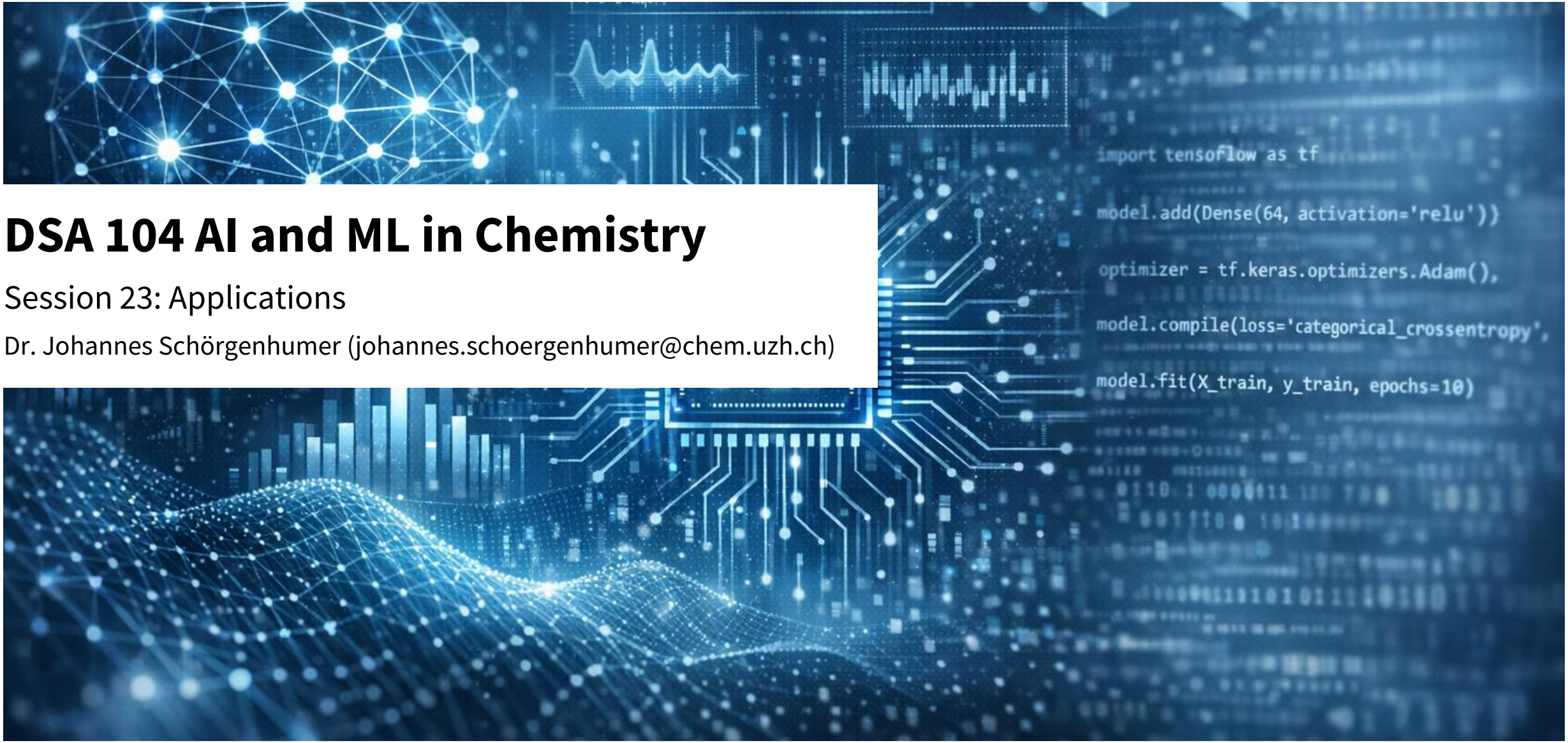
Universität Zürich[UZH]

Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 23: Applications

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Dr. Stephan Guttinger, University of Exeter (UK):

The Ethics of Automated Research

DSA 104 – Virtual guest lecture

What are some core ethical issues with AI and Automation?

What does research ethics look like when experiments are planned, performed and analysed by multi-agent AI systems instead of humans?

What is the role of the human in automated research?

Join Dr. Stephan Guttinger's guest lecture to find out!

When?

May 13, 13:00 – 14:45

Where?

Hybrid lecture: **Y19-J-94**

MS Teams: Scan the QR Code

DSA Symposium - May 27, Y11-F-06

Agenda and topics (Booklet and official announcement to follow):

08:00 - 08:30 DSA102 Wrap up and feedback

DSA 102 Session - Modern lab technologies:
08:30 - 09:15 Tina Binz (UZH): *Advancing hair and nail matrix analysis with automated workflow solutions*
09:15 - 10:00 Glenn Bojanov (Uni Bern): *Biocatalysis as a Tool in Synthesis: Continuous-Flow Routes to Greener Chemistry*

10:00 - 10:30 Coffee break

10:30 - 11:15 Tristan Cofer (Uni Bern): *title tbd* (Topic: Plant sciences – biotic interactions)

11:15 - 12:00 Joshua W. Sin (Roche): *Optimising Chemical Process Development with Automation and Machine Learning: An Industrial Perspective*

12:00 - 13:00 **Lunch** with Speakers and registered participants (Y38, K floor, register via mail)

DSA 104 Session – AI and ML in the life sciences:
13:00 – 13:40 Christoph Schür (EAWAG): *Machine learning for predictive ecotoxicology*
13:40 – 14:20 Sarina Kopf (EPFL): *Generative machine learning for catalyst discovery*
14:20 – 15:00 Loïc Roch (Atinary): *title tbd* (Topic: self driving labs)

15:00 – 16:00 Apéro

Applications of AI in the life sciences

Historic perspective

Application scenarios in literature

Literature discussion

— 2 Examples

— Discussions and presentations of assignment 8

— Closer look at 2 selected papers of assignment 8

Historic perspectives

Many concepts have been developed very early on (e.g. MLP, Bayesian optimisation, …)

Broader adoption in life sciences came much later (after 2012)

With DL: long-standing challenges solved, applications diversified massively, e.g.

— Powerful task-specific predictions/generations

— Scientific writing

— Automated Laboratories (SDLs)

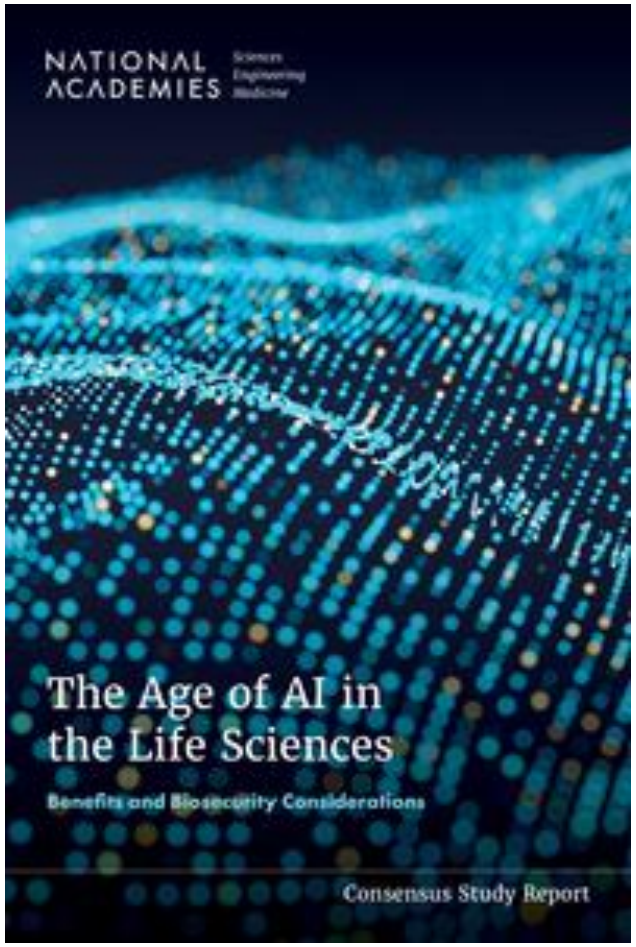
— Enhanced decision making

But: new challenges arise, e.g.

— Missing regulations

— Misuse

— Lack of validation



The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington (DC): National Academies Press (US); 2025 Apr 23. Available from: https://www.ncbi.nlm.nih.gov/books/NBK614593/ doi: 10.17226/28868

Historic perspectives in literature

Literature count (Web of Science, …)…

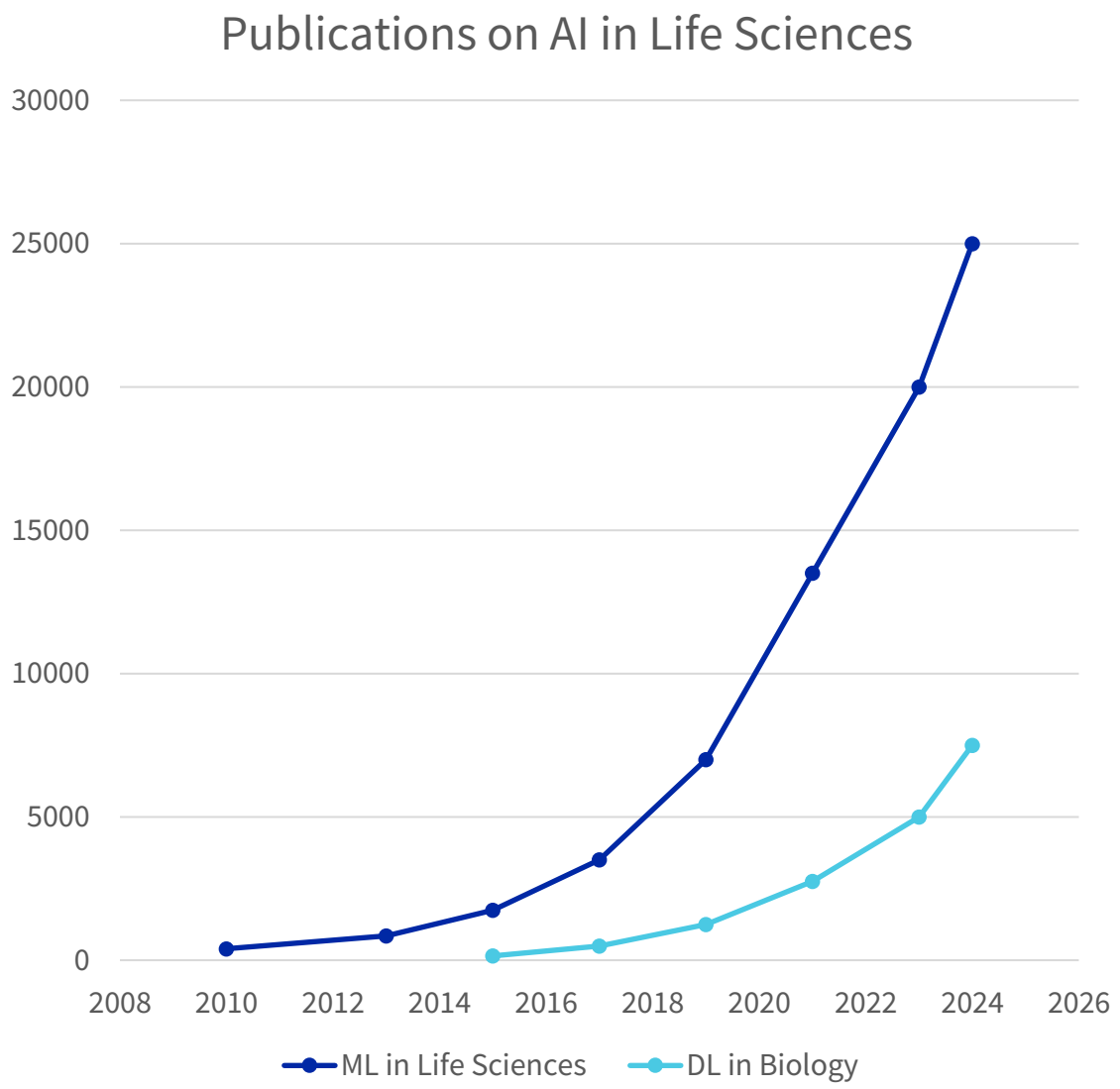
— Approximate

— Clear trend: Explosion around 2016

Compare:

— DL in Biology: >7000 papers in 2024

— DL globally: ~60000 publications in 2024



Applications

E.g.

— Structure prediction

— Sequence generation / enzyme design

— Metabolite discovery

— Synthesis planning

— Accelerated drug discovery

— Reaction condition optimisation

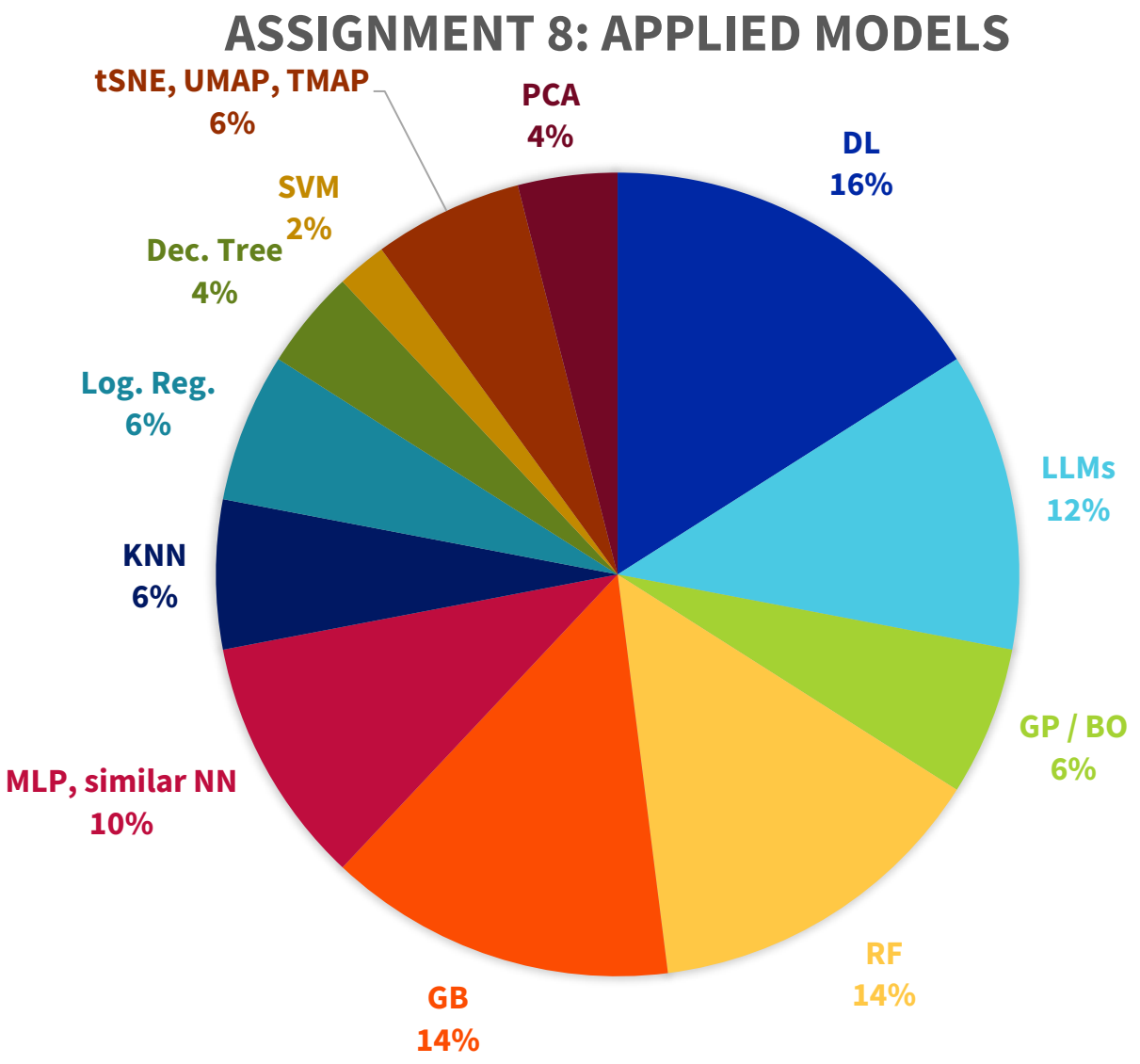
— Medical condition assessment

— Automation (SDLs, etc.)

— Analytics

— …

In short: everywhere.



Exploration in literature – limitations in reaction data

Grzybowski, Burke et. al.: **Machine Learning May Sometimes Simply Capture Literature Popularity Trends: A Case Study of Heterocyclic Suzuki–Miyaura Coupling**

Goal: Predict reaction conditions for Suzuki couplings

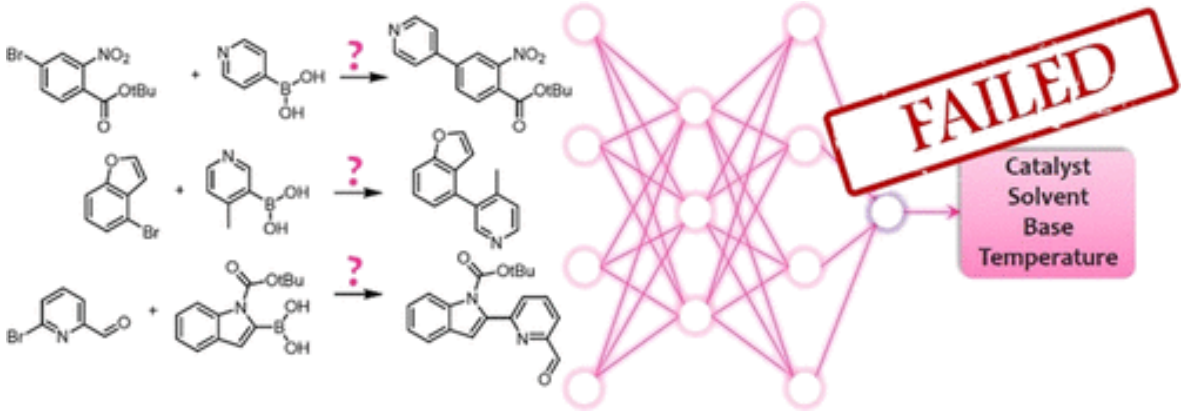
— **Rich data**, carefully curated (Reaxys, >13000 Suzuki couplings, USPTO)

— **Feature engineering**: descriptors, fingerprints

— NN **failed**

— GCNN **failed**, even with restricted search space (solvent, base)

— Literature-known DL models **failed**



— Conclusion:

  — **Failed = similar to "popularity baseline"**

  — "Chemistry propagates its own routines": **Conditions unusable as descriptors, incorporate little variance**

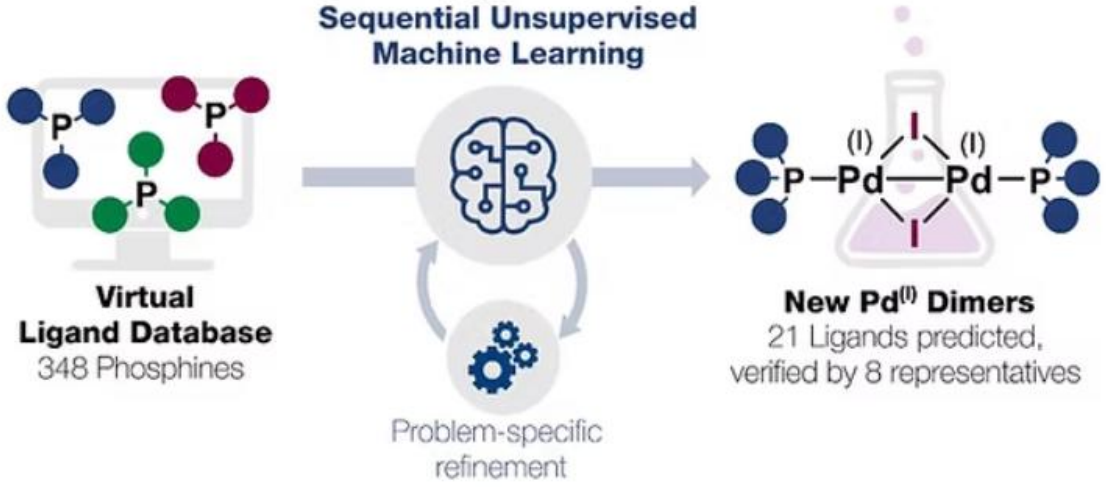
Journal of the American Chemical Society **2022**, *144*, 4819-4827; DOI: 10.1021/jacs.1c12005

Exploration in literature – new catalysts by unsupervised ML

Schoenebeck et. al.: **Accelerated dinuclear palladium catalyst identification through unsupervised machine learning**

Goal: Stabilise Pd dimers (catalytically active species)

- Sparse data regime (data = 348 ligands)
- Simple recipe (k-means with PCA visualisation):
  - Clustering based on general descriptors (reduce to 89 ligands)
  - Derive problem specific descriptors (DFT)
  - Problem-specific clustering
  - Candidate selection based on known data



- Validation:
  - Average clustering (1000 iterations)
  - 21 ligands experimentally tested, 8 new complexes found.

Science **374**, 1134-1140 (2021). DOI:10.1126/science.abj0999

Literature deep-dive

In assignment 8, some contemporary literature has been collected and can be roughly distributed into these categories

— Omics: Paper 9, paper 11

— Enzymes, Proteins (Sequence Learning): 8, 10, 12, 13

— Drug development: 1, 16, 17

— Medical assessments: 7, 18

— Reaction opt: 2, 3, 4, 5, 14, 23

— Retrosynthesis, react. pred., chemical space: 2, 6, 15, 19, 24

— Analytics: 22

— Reviews: 7, 20, 21

1. Get together in small groups and each picks a paper (topic diversity! Do not pick a review and not paper 2).
2. This paper should then be discussed in detail in the group and the most important highlights should be distilled.
3. Present the content of the paper and **try to convince** a critical audience (= the class ☺)
4. In a more open discussion we will focus on limitations, biases, …

Literature deep-dive

1. Get together in small groups and each picks a paper (diversity!).
2. This paper should then be discussed in detail in the group and the most important highlights should be distilled.
3. Besides general points, discuss these points:
   - What question/task is being answered/solved computationally?
   - Where does learning replace explicit modeling?
   - What is the ground truth and how reliable is it?
   - What might break when the model is deployed outside the paper?
   - What new kinds of errors or biases does ML introduce?
4. Present the content of the paper and **try to convince** a critical audience (= the class ☺)
5. In a more open discussion we will focus on limitations, biases, …

Assignment 8 – a closer look at paper19

Have a look at the review *paper19* and discuss:

— For the illustrated examples in Fig 4, 9 and 13: What do the numbers mean for you?

— What kind of algorithms were used?

— What data?

Assignment 8 – a closer look at paper2

Investigate *paper2* and discuss:

— What role do the agents fulfil?

— What advantages does ChemCrow offer?

— How would you compare that to single task-specific models?

Application – broader discussion

— AI omnipresent, also in life sciences, all fields of applications.

Discussion points:

— What are some considerations? Critical questions that you should always pose?

— Limitations?

— Still discoveries to be made?

— How should you conduct research in the AI age?

On Policies:
https://www.ai.uzh.ch/en/research/ai-policies.html
https://www.mnf.uzh.ch/en/studium/rund-ums-studium/Studium_und_KI.html